Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MD Razia
Student Name	Ritwik
Roll Number	2520080051
Branch	AI&DS
Course Outcome	Co-3

WaterGrid – Smart Water Distribution & Monitoring System

Work Done Summary:

The WaterGrid Smart Water Distribution & Monitoring System was successfully implemented using Kruskal's Minimum Spanning Tree algorithm. The MST connected all water distribution hubs with the minimum pipeline installation cost while avoiding redundant loops. The optimized network achieved a minimum total pipeline cost of 9 units, demonstrating an efficient city-wide water network.

Implementation Details:

1. Represented water distribution hubs as graph vertices in the WaterGrid network.

2. Represented pipeline connections as weighted edges with installation cost.

3. Applied Kruskal's Algorithm to construct the Minimum Spanning Tree (MST).

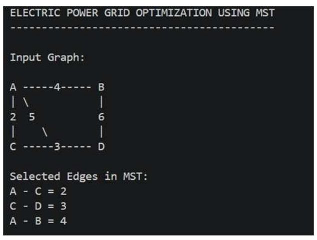
4. Sorted pipeline edges based on installation and maintenance cost.

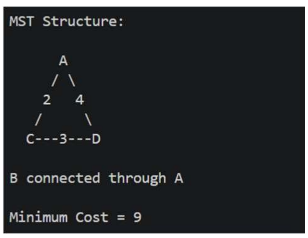
5. Used Union-Find for cycle detection to prevent redundant pipeline loops.

6. Connected all water distribution hubs with minimum total pipeline cost.

7. Calculated the total optimized WaterGrid pipeline network installation cost.

Results and Screenshots:





In this case study, we implemented WaterGrid, a Smart Water Distribution & Monitoring System using the Minimum Spanning Tree (MST) concept. Water distribution hubs are represented as graph vertices, and pipeline connections are represented as weighted edges. Kruskal's Algorithm was used to connect all water hubs with the minimum pipeline installation cost while avoiding redundant loops. The MST ensures an optimized and cost-effective water distribution network.

GitHub Link: https://github.com/ritwikxdevv/DSA-CASE-STUDIES

Student Signature	Faculty Signature